

PID Controllers

Industrial Control Systems · SCADA Hub

COURSE DESCRIPTION

PID controllers are the primary feedback control element in industrial process automation, governing temperature, pressure, flow, and level loops in every manufacturing and utility sector. This course covers the proportional, integral, and derivative terms; process characterization via step testing; classical and lambda tuning methods; cascade control design; and PID function block configuration across major PLC platforms.

PREREQUISITES

Basic algebra. No prior control theory or PLC programming experience required.

COURSE TOPICS

1. **Introduction:** Feedback control concept and the uniform mathematical structure underlying temperature, pressure, flow, and level applications.
2. **Control Loop Fundamentals:** Process variable, setpoint, error, and manipulated variable — vocabulary prerequisite for PID algorithm study.
3. **PID Algorithm:** Proportional, integral, and derivative terms; the specific problem each term solves; combined three-term controller behavior.
4. **Tuning Methods:** Ziegler-Nichols closed-loop and open-loop, lambda tuning, IMC; method selection criteria and failure modes.
5. **Process Characterization:** First-order plus dead time model, step test procedure, and extracting gain, time constant, and dead time from equipment.
6. **Cascade Control:** Inner and outer loop design, commissioning sequence, and disturbance rejection improvement over single-loop control.
7. **Digital Implementation:** Sampling rate, velocity-form vs. position-form algorithm, bumpless transfer, and anti-windup on saturated outputs.
8. **PLC Implementation:** PID function block configuration across Rockwell, Siemens, Beckhoff, and Modicon; scaling conventions; commissioning procedure.
9. **Troubleshooting:** Oscillation, offset, slow response, and derivative chatter — each symptom mapped to a specific tuning adjustment.
10. **Practice Lab:** Flow loop step-test tuning, cascade temperature/flow loop, anti-windup configuration, cold-start commissioning sequence.

LEARNING OUTCOMES

- Perform a step test on live equipment and extract first-order plus dead time model parameters.
- Apply Ziegler-Nichols and lambda tuning methods to self-regulating process loops.
- Design and commission a cascade control loop with sequenced inner and outer tuning.
- Configure anti-windup and bumpless transfer on a PLC PID function block.
- Diagnose oscillation, steady-state offset, and derivative chatter from trend data alone.

REQUIRED SOFTWARE

- **Scilab** (free, cross-platform) — numerical computing environment for step-test simulation and response plotting.
- **CODESYS** (free, Windows/Linux) — soft PLC runtime for PID function block configuration and commissioning exercises.